Advances in Brain Injury Research and Care in China 1

Stroke in China: advances in prevention and management on the path to a healthy China

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The incidence and mortality of stroke have decreased in China in the past decade; however, the prevalence of stroke continues to rise because of population ageing. In alignment with the Healthy China 2030 blueprint, there has been a strategic reorientation of health-care services towards the prevention of major diseases, including stroke. Along with the establishment of a nationwide network of stroke centres to enhance the delivery of stroke care and research, progress has also been made in understanding stroke aetiology, screening for and control of risk factors, and promoting early diagnosis and treatment. Evidence from high-quality randomised trials in China supports the wide adoption of reperfusion therapies for ischaemic stroke, early blood pressure lowering after intracerebral haemorrhage, individualised approaches to secondary prevention, and the research evaluation of traditional Chinese medicine and neuroprotective agents. Future efforts should focus on promoting a healthy lifestyle, public education on awareness of stroke prevention and timely access to stroke services, training of stroke clinicians on evidence-based stroke care, and improvements in both pre-hospital and post-hospital stroke services.

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This is the first in a **Series** of two papers on Advances in Brain Injury Research and Care in China

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Introduction

Globally, more than 100 million people have a stroke,[1] and according to the Global Burden of Diseases (GBD) Study 2023, China has the highest prevalence of stroke in the world.[2] Furthermore, stroke accounts for the highest disability-adjusted life-years (DALYs) of 371 diseases and injuries in China.[3] The burden of stroke is expected to continue rising in relation to population ageing. To address the health challenge of stroke and other major diseases, the Chinese government launched the Healthy China 2030 blueprint in 2016, declaring public health a priority for national socioeconomic development.[4] The blueprint includes five goals to: (1) enhance overall health promotion, (2) control major risk factors, (3) improve the capacity and quality of health services, (4) enlarge the scale of health industries, and (5) upgrade health systems and policies. To achieve these goals in relation to stroke care, a series of governmental initiatives have been implemented, which in the context of rapid economic development and wide implementation of evidence-based medicine, propel the progress in stroke care in China.

In previous reviews, we summarised advances in clinical practice and research in stroke care.[5,6] In this first paper of a Series on Advances in brain injury research and care in China, we now provide an update on: the epidemiology of stroke; the current state of stroke services; and progress in stroke prevention, treatment, and research. Our focus is on advances in understanding aetiology, management of risk factors, and capacity development to improve the quality of stroke services, including the provision of reperfusion therapies. We also discuss the progress and gaps in low-resource settings, and propose directions for future research and the enhancement of clinical practice.

The burden of stroke in China

China has endeavoured to reduce the burden of stroke in the past decades with some staged success. According to GBD 2023 estimates, the age-standardised incidence, mortality, and DALYs of stroke in China have declined from 1990 to 2023, consistent with global trends in countries with a different sociodemographic index.[2] However, in the past 30 years, the age-standardised prevalence of stroke in China has fluctuated, with a generally stable trend for an increased prevalence of ischaemic stroke, which contrasts with the decreasing global trend (appendix pp 12–13). The latest GBD data revealed a high burden of stroke in China, with estimates of 26·7 million prevalent cases, 4·2 million incident cases, and 2·1 million stroke-related deaths in 2023.[2] However, there is no nationwide survey designed for stroke epidemiology in China to provide updated granular data since 2013.[6]

The declining trend in the overall incidence of stroke in China is consistent with decreases in the incidence of both intracerebral haemorrhage and subarachnoid haemorrhage, despite an increase in the incidence of ischaemic stroke (appendix pp 12–13). Globally, in 2021, the incidence of ischaemic stroke was more than twice that of intracerebral haemorrhage, yet intracerebral haemorrhage was associated with similar numbers of deaths and more DALYs than ischaemic stroke; moreover, China had the largest burden of intracerebral haemorrhage-related DALYs in the world.[7] These epidemiological estimates highlight the imperative need to prevent and to improve the management of both ischaemic stroke and intracerebral haemorrhage.

According to the National Health Service Surveys in China, the crude prevalence of stroke in rural areas

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See Online for appendix

surpassed that in urban areas in 2013, and the discrepancy became more evident in 2018 (2700 per 100 000 people in rural areas *vs* 2000 per 100 000 people in urban areas in 2018; figure 1).[8] Based on statistics of the National Health Commission, stroke ranked as the third leading cause of death in both urban and rural areas in 2021, following heart disease and cancer.[9] Age-standardised mortality of stroke has fluctuated in the past 20 years, with an overall decreasing trend (from 167 to 131 per 100 000 person-years between 2003 and 2021), but it is consistently higher in rural than urban areas (figure 1). The greater burden of stroke in rural areas might be associated with relatively limited health-care resources and the ageing of the rural population, as the younger population migrates to cities (appendix p 14).

A previous nationwide survey revealed regional disparities in stroke burden in China, with a higher mortality-to-incidence ratio of stroke in the southwest region compared with other geographical areas.[6] The 2019 GBD study provided more detailed data to estimate the mortality-to-incidence distribution, which was highest in both Xizang and Qinghai, in western China,

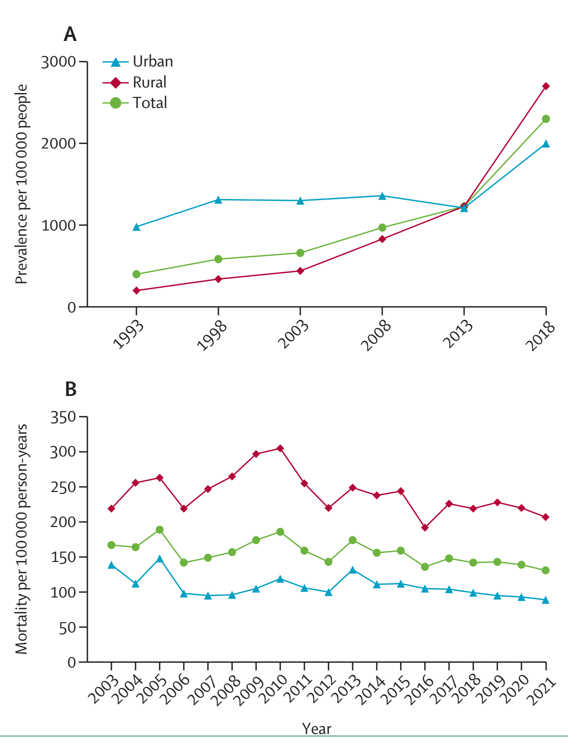

Figure 1: **Prevalence and mortality of stroke in China**
(A) Crude prevalence of stroke in rural and urban populations in China (per 100 000 people). Data were extracted from the quinquennially published *Report of the National Health Service* for 1993–2018. The prevalence data in 1993, 1998, 2003, and 2008 were derived from all age groups and the prevalence data in 2013 and 2018 were derived from the population aged 15 years or older.[8]
(B) Age-standardised mortality of stroke in rural and urban populations in China (standardised by the China 2020 census population, per 100 000 person-years). Data were extracted from the annually published *China's Public Health Statistical Yearbooks* for 2003–21. Each yearbook reports data of the preceding year.[9]

and lowest in Macao (the southeast costal region; figure 2).[10] The distribution of mortality-to-incidence reflects both accessibility and quality of stroke services, which presents an inverse trend to the distribution of health-care resources: compared with eastern China, western provinces and regions have generally fewer registered physicians per 1000 residents (figure 2), and fewer secondary and tertiary hospitals per 10 000 km² territory (figure 2).[9]

Stroke aetiologies

Reliable assessment of aetiologies and pathological subtypes of stroke can advance our understanding of patterns of disease, inform strategies for prevention and treatment, and allow the allocation of health-care resources to be optimised. According to GBD 2023 estimates, ischaemic stroke accounted for 62·8%, intracerebral haemorrhage for 32·6%, and subarachnoid haemorrhage for 4·6% of all incident strokes in China.[2] Cerebral venous thrombosis accounts for 0·5–1·0% of all strokes in other countries;[12] however, epidemiological data in China remain limited.[13]

Large artery atherosclerosis is the major aetiology of ischaemic stroke in China.[14,15] A high proportion of large artery atherosclerosis-related stroke is associated with a higher prevalence of intracranial atherosclerosis in Chinese than in Western populations.[16] A single-centre longitudinal study reported a decline in age-standardised incidence of ischaemic stroke and in large artery atherosclerosis-related stroke in China from 2004 to 2018.[17] The reduction in stroke burden in this cohort was accompanied with a decreasing proportion of large artery atherosclerosis-related stroke, from 23% of ischaemic stroke events in 2004 to 9% in 2018, of which the 2018 proportion was similar to that observed in concurrent studies in Western countries (8–9%, appendix p 18). This finding was probably related to the increased detection of cardioembolic stroke and small artery occlusion during the study period, and it might also reflect the effective control of vascular risk factors that was implemented in the local healthcare system. However, large artery atherosclerosis remains a predominant stroke aetiology within the broader Chinese population (22–45% of all ischaemic stroke cases; appendix p 18). It should be noted that the reported proportions were not directly compared between different regions within China nor between China and Western countries. Despite this fact, it is likely that more effective control of vascular risk factors could help to narrow any disparity.

The proportion of stroke related to cardioembolism might be lower in Chinese patients than in those from Western countries. Since the prevalence of atrial fibrillation rises with age, the low proportion of cardioembolism in Chinese patients can be partly explained by their younger average age than patients in Western countries (mean age 65 years *vs* older than age 70 years; appendix p 18). Discrepancies in the reported proportions of cardioembolism might also reflect variation in the completeness of evaluations and in the expertise of assessors, as indicated by a study showing that the central adjudication of aetiology subtyping shifted more cases to be cardioembolism than when aetiology was reported by site neurologists.[15] Prolonged

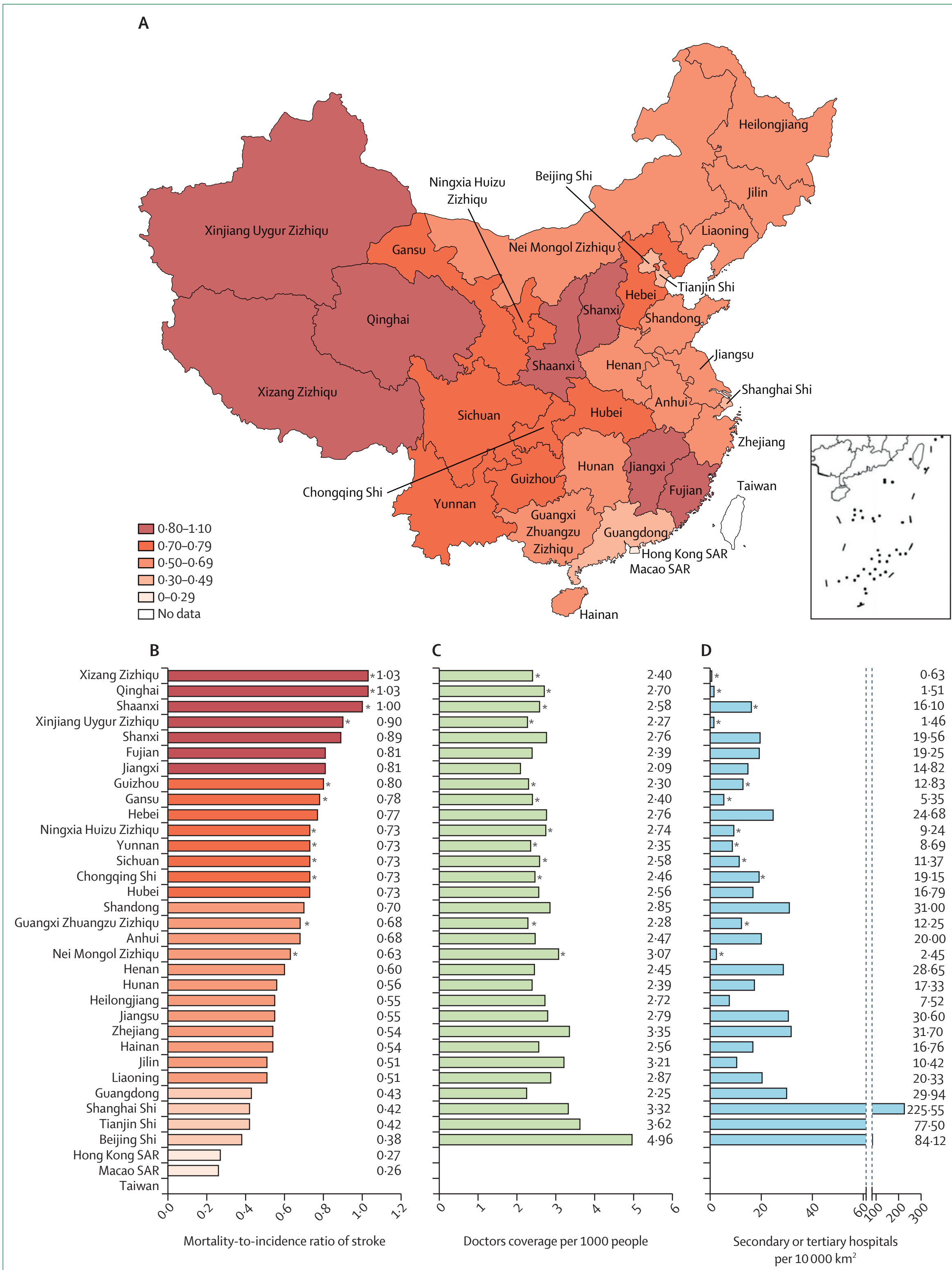

***Figure 2:* Provincial distribution of mortality-to-incidence ratio of stroke and health-care resources in China**

(A) Geographical distribution of mortality-to-incidence ratios of stroke in China.[10] (B) Details of mortality-to-incidence ratios of stroke in 34 provinces, autonomous regions, municipalities, and special administrative regions, calculated from the Global Burden of Disease Study 2019.[10] (C) Geographical distribution of the number of registered physicians per 1000 people. The number of registered medical or associate health-care professionals, and the population in each region were retrieved from the *China's Public Health Statistical Yearbooks 2023*.[9] (D) Secondary or tertiary hospitals per 10 000 km². The number of secondary or tertiary hospitals was retrieved also from the *China's Public Health Statistical Yearbooks 2023*.[9] The area of the territory in each region was retrieved from the website of the Ministry of Civil Affairs of the Peoples' Republic of China.[11] Data are not available for the Taiwan region (figure 2A–D), Hong Kong SAR (figure 2C–D), and Macao SAR (figure 2C–D). SAR=special administrative region. *Regions in western China.

cardiac rhythm monitoring can lead to higher rates of detection of atrial fibrillation,[18] and it has been increasingly used in clinical practice in China in the care of patients with suspected cardiogenic or cryptogenic stroke.

Small vessel disease is the most frequent cause of intracerebral haemorrhage.[19] A study of 1938 Chinese patients with intracerebral haemorrhage between 2012 and 2018 identified hypertensive angiopathy as the most common subtype (66·8% of all events) followed by cerebral amyloid angiopathy (12·9%).[20] Small vessel disease also causes ischaemic stroke, known as lacunar infarction or small artery occlusion, which accounts for about 30–40% of all ischaemic strokes in China (appendix p 18). In Chinese patients, approximately one third of lacunar infarctions identified on brain imaging appear to be clinically silent. Asymptomatic lacunar infarctions result in similar risks of recurrent stroke and mortality as symptomatic lacunar infarction.[21] Yet, there is little evidence to inform the treatment of silent lacunar infarction.[22] Moreover, heterogeneous pathophysiological mechanisms underlying lacunar infarction complicate its management.[23] High-resolution MRI can facilitate differential diagnosis of aetiologies for lacunar infarction.[24]

Risk factors and primary prevention

Stroke is an eminently preventable condition, since a high proportion of events could be attributed to modifiable factors. Hypertension is the top risk factor for stroke in China.[25] A community-based national survey in 2018 estimated 274 million prevalent cases of hypertension in Chinese adults. The age-standardised prevalence was 20·8% in 2004 and 24·7% in 2018; meanwhile, disease awareness increased from 31% to 38% among those who are hypertensive, treatment of hypertension from 26% to 35%, and hypertension control from 7% to 12%.[26] The general improvement in hypertension detection and management might reflect the successful implementation of national public health programmes to screen for and control hypertension.[27] However, hypertension awareness, treatment, and control appear much lower than those in high-income countries.[28] The estimated prevalence of dyslipidaemia (34% of the adult population aged between 35 and 75 years)[29] and diabetes (about 12% of adults aged 18 years or older)[30] have remained stable within the past decade.[6] In general, management of these metabolic risk factors was better in people living in urban areas than in rural areas, which emphasises the importance of public-health educational campaigns and health-care provision targeting rural populations.[26,29,30] Compared with non-invasive blood pressure monitoring approaches, the cost and discomfort associated with blood sampling to measure blood lipids and glucose are barriers to widespread screening and monitoring of dyslipidaemia and diabetes. Hence, low-cost and more acceptable devices are needed to promote monitoring of blood lipids and glucose.

Dietary risk factors are estimated to be responsible for 16 million stroke-related DALYs in China, with geographical disparities, as stroke-related DALYs are fewer in provinces with better economic development.[31] A low intake of fruit is the leading dietary risk factor for stroke, followed by low intakes of whole grains and vegetables.[31] According to the 2018 China National Nutrition Survey, the mean consumption of vegetables (390 g per day) matched guideline recommendations, while the consumption of fruits (114 g per day) and whole grains (21 g per day) remained suboptimal, although their consumption had increased since 2002.[31] The National Health Commission has recently issued a recommendation for Chinese adults, with the daily consumption of 300 g of vegetables, 200–350 g of fruit and 50–100 g of whole grains, and 300–500 g of aquatic products (eg, fish, shrimp, shellfish, etc) per week.[32] There were geographical disparities in diet-related stroke mortality rates.[31] Furthermore, geographical variations in culture and food supply challenge efforts to modify dietary patterns.[33] The EAT-*Lancet* Commission recommends individualised energy consumption that consists of high consumption of fruit and vegetables, low-to-moderate intake of seafood and poultry, and low consumption of red meat and processed food.[34] This framework can provide a health reference diet for adoption within different geographical regions and food cultures in China.

Alcohol consumption is a traditional social custom in China. A nationwide survey revealed that one third of Chinese men were regular consumers,[35] despite the total alcohol consumption per person per year decreasing from 7·5 L in 2015 to 4·5 L in 2022.[36] A Mendelian randomisation study examined the role of east Asian-specific genetic variants that influence alcohol metabolism, which indicated that moderate alcohol drinking increased the risk of stroke, refuting the reported beneficial effects of moderate alcohol intake shown in conventional epidemiological studies.[37] China is the largest consumer of tobacco in the world.[38] Tobacco smoking is associated with prevalence and mortality of many non-communicable diseases, including stroke.[39] There was a slight decrease in the prevalence of smoking in adult men, from 58% in 2007 to 51% in 2018, and the decrease was more prominent in urban than rural areas, whereas a smoking prevalence of 2% was unchanged in women.[40] To diminish the adverse health effects of tobacco smoking, the Chinese government ratified the WHO Framework Convention on Tobacco Control in 2005, which imposed regulations on tobacco tax, package warnings, restrictions on tobacco advertising, public smoking bans, and implementation of smoking cessation services.[41] With ongoing challenges, there is hope to achieve the national target of reducing the overall smoking prevalence to 20% of the population aged 15 years or older by 2030 (currently 22·9% in 2025[42]).

WHO recommends that adults undertake physical activity, either at moderate intensity for at least 150 min or at vigorous intensity for at least 75 min per week, to maintain cardiovascular fitness.[43] In China, domestic and occupational physical activity has declined due to widespread changes towards urbanisation. In a survey involving 4 229 616 community-dwelling Chinese residents, the age-standardised prevalence of physical inactivity, defined as a failure to meet the WHO recommendation, increased from 22% to 29% between 2013 and 2019, which corresponded to an increased risk of stroke.[44] The popularisation of public bike sharing schemes in many urban and rural areas offers an environment-friendly commuting option to boost physical activity during commuting and leisure time.[45] The schemes emerged in 2016 and rapidly expanded to an estimated 12 million bikes offering 27 million daily rides in 2024.[46]

A downside to outdoor activities is exposure to air pollution, which can counteract the protective effects of physical activity.[47] Pollution due to particulate matter (PM), lead exposure, and household air pollution from solid fuels are major environmental factors that contribute to stroke burden in China.[25] Long-term exposure to high concentrations of fine PM of diameters 2·5 μM or smaller ($PM_{2·5}$) is associated with an increased risk of incident stroke.[48] Lead exposure, mainly from diet and air, is also associated with stroke-related DALYs.[49] China launched a clean air action plan in 2013, which reinforces major control measures to manage industrial and agricultural sources of pollution, promote the residential use of clean fuels, and control emissions from motor vehicles.[50] The first two phases of the action plan have been associated with a substantial reduction in $PM_{2·5}$ concentrations, while the third phase is ongoing.[51]

Lifestyle modification is a cost-effective and widely accessible approach for the prevention of non-communicable diseases.[52] Given the multidimensional nature of stroke risk factors, the Healthy China action plan encompasses a multifaceted approach at both public-health and individual levels to manage vascular risk factors, such as hypertension, dyslipidaemia, and diabetes, and to foster behavioural changes in diet, smoking, alcohol intake, and physical activity (figure 3).[54] Tools that integrate these multidimensional factors might help doctors to more efficiently assess and monitor risk factors, thus, improving brain health and reducing stroke risk.[55]

Compared with urban China, rural areas have the higher burden of stroke and vascular risk factors but less health-care resources. Population-based interventions have been developed to address the pressing need for cost-effective stroke prevention in rural areas. The Salt Substitute and Stroke Study (SSaSS) found that the use of a potassium-substitute reduced-sodium salt for cooking was feasible in rural China, and compared with regular salt, it reduced systolic blood pressure and the risk of stroke within 5 years of follow-up.[56] In urban areas, specialists and primary care physicians work collaboratively to manage vascular risk factors in the community, but in many rural areas, village doctors (ie, non-physician community health-care providers)[57] are often the first-line health-care providers. The China Rural Hypertension Control Project (CRHCP) showed that an intensive blood pressure intervention programme delivered by trained and supervised village doctors improved blood pressure control[57] and reduced the incidence of cardiovascular diseases compared with standard care.[58] A system-integrated technology-enabled intervention (SINEMA) showed that the trained village doctor in rural China equipped with digital health technology was effective in reducing blood pressure and improving secondary prevention in people who have had stroke.[59] These findings highlight the great potential of leveraging the existing health-care system for improving prevention strategies in low-resourced rural settings, by training health-care professionals and using digital and telemedicine technologies.

Health-care services for stroke

National policy and strategies

To achieve better overall health, including the reduction of the burden of stroke, the State Council launched the Healthy China 2030 blueprint in 2016.[4] The blueprint highlights the importance of a lifecycle strategic reorientation from treatment to prevention for major chronic diseases.[60] In alignment with Healthy China 2030, the National Health Commission launched the Million Disability Reduction Project to strengthen stroke prevention by screening for vascular risk factors, and to improve the capacity and quality of stroke services by implementing a national network of stroke centres.[61] There has been an overall improvement in the quality of stroke care in the past 20 years, as shown by data from the nationwide stroke registries (table 1).[62–73] In general, stroke care involving widely accessible medications, such as anti-platelet and lipid-lowering drugs, has improved greatly and now about 90% of hospitalised patients with stroke will have access to these treatments; however, stroke care involving advanced technologies and costly medications (eg, reperfusion therapies) is progressing slowly. In high-income regions, such as Eastern China and urban areas, the quality of stroke care is better than in other regions.[64,74] To promote evidence-based stroke care across the country, the Chinese Stroke Society, the Chinese Stroke Association, and other academic stroke societies (appendix p 19) have issued and regularly updated a series of stroke guidelines.[75,76] Furthermore,

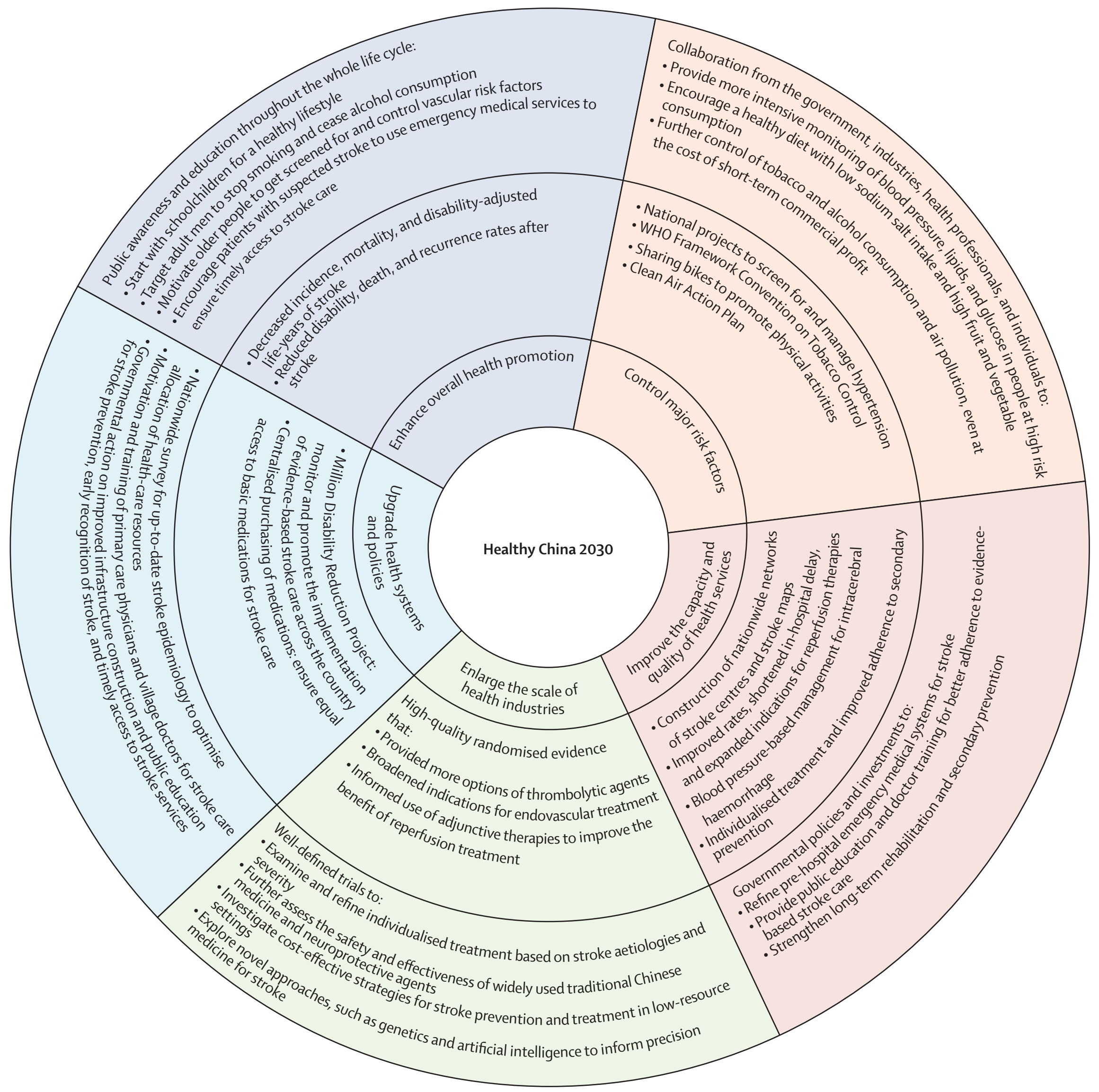

Figure 3: **Progress and gaps in stroke care on the path to a healthy China**
Inner circle: the five goals of the Healthy China 2030 blueprint;[53] middle circle: progress in stroke care in China; and outer circle: gaps in stroke management and directions for future research.

several national and regional interventional programmes have been designed to further improve the adherence of clinicians to evidence-based stroke care (appendix pp 20–23).

Reperfusion therapies for acute ischaemic stroke

Delays in treatment after the onset of stroke symptoms can lead to severe damage from brain ischaemia and compromise the benefit of reperfusion treatment. Recognition of these harmful effects of treatment delays has stimulated efforts to raise stroke awareness, reduce delays in diagnosis, and ensure prompt treatment. The establishment of green channels (defined as dedicated stroke pathways implemented by multidisciplinary health providers to optimise workflow efficiency in hyperacute stroke care) and standard operating procedures for emergency stroke care has reduced in-hospital delays, with the door-to-needle time shortened from a median of 94 min in 2008 to 60 min in 2018.[66] Despite an improvement of in-hospital time parameters, reducing the delay in the pre-hospital phase has been less successful. A national network of stroke centres and regional networks of stroke rescue maps have been created to ensure the 1-h stroke rescue circle for emergency care. Mobile stroke units have also been tested in some regions.[77] Furthermore, various community educational programmes have been conducted to improve public awareness about early signs of stroke and the importance of timely arrival to hospital, via television channels, newspapers, and digital and social media.[78,79]

Considerable progress has been accomplished in the promotion of reperfusion therapies. Intravenous alteplase, the first medical therapy approved for the acute treatment of patients with ischaemic stroke in China, has a strict therapeutic time window of within 4·5 h after symptoms onset. Although the nationwide rate of intravenous thrombolysis has improved from around 10% to 30% of eligible patients in the past decade (table 1), only one third of patients with acute ischaemic stroke get admitted within the therapeutic time window.[70] This statistic reinforces the great need to expand indications for intravenous thrombolysis. The current Chinese stroke guidelines recommend the use of alteplase or tenecteplase, with similar indications and contraindications, within 4·5 h of symptoms onset, and the use of urokinase within 6 h of symptoms onset.[75] The efficacy of alternative thrombolytic agents, such as reteplase and prourokinase, is being assessed in clinical trials, and new studies have emerged to support an extended therapeutic window for intravenous thrombolysis of up to 24 h after a stroke in selected patients (appendix pp 24–40). More studies are needed to develop novel technologies, such as artificial intelligence, to identify individuals with salvageable brain tissue and thus, extend the benefits of reperfusion therapies to a broader population.

The benefit of endovascular mechanical thrombectomy for patients with acute ischaemic stroke from large vessel occlusion in anterior circulation was established about a decade ago.[80] The number of patients with large vessel occlusion who received thrombectomy has increased by more than ten-fold, from less than 3000 patients per year before 2015 to more than 40 000 in 2020.[81] However, this latter number only accounts for approximately 1·5% of patients with acute ischaemic stroke (table 1). This small proportion was related to restricted indications, high costs, limited infrastructure, and the need for highly skilled health-care professionals and neuro-interventionists. To extend the access to endovascular treatment, the Chinese government has supported the construction of stroke centres across the country. As of December 2025, there are 1698 accredited secondary stroke centres (ie, capable of emergency stroke care and thrombolysis) and 769 tertiary stroke centres (ie, focused on advanced stroke care, such as multimodal imaging and endovascular treatment), accounting for all provinces and about half of all counties in mainland China.[82] In stroke centres with the capability for thrombectomy, the intervention is accessible 24 h and 7 days a week. Furthermore, regional networks of endovascular treatment have been developed based on telemedicine and mobile technologies, in which tertiary stroke centres collaborate with emergency medical services and secondary stroke centres.

Evidence derived from trials done in China, along with accumulating evidence from international trials, has expanded indications for thrombectomy, which now include patients with large infarction in the anterior circulation and patients with basilar artery occlusion, with the therapeutic window extended from 6 to 24 h for selected patients. Evidence from clinical trials in China suggest that direct mechanical thrombectomy is non-inferior to intravenous alteplase bridging to mechanical thrombectomy, while intravenous tenecteplase before thrombectomy is superior to direct thrombectomy. The evidence does not support intra-arterial thrombolysis after a successful reperfusion (appendix pp 24–40).

Management of haemorrhagic stroke

The high burden of disability and mortality due to intracerebral haemorrhage is problematic and requires solutions. The control of high blood pressure is a mainstay of the treatment of the acute phase of intracerebral haemorrhage. The third Intensive Care Bundle with Blood Pressure Reduction in Acute Cerebral Haemorrhage Trial (INTERACT3), which involved a large proportion of participants from China, established the benefit of improved systems of care that involve a care bundle incorporating early intensive blood pressure lowering alongside the management for hyperglycaemia, pyrexia, and abnormal anticoagulation on functional outcome after intracerebral haemorrhage.[83] This landmark trial has accelerated interest in the use of acute care bundles for the management of intracerebral haemorrhage, worldwide.[84] A subgroup analysis of the fourth Intensive Ambulance-Delivered Blood Pressure Reduction in Hyper-Acute Stroke Trial (INTERACT4) further confirmed the benefit of intensive blood-pressure lowering initiated in the ambulance within the first 2 h after the onset of symptoms of intracerebral haemorrhage.[85] Conversely, however, findings from INTERACT4 also showed that early intensive blood-pressure lowering was harmful in patients with ischaemic stroke, with subsequent Chinese trials further indicating that intensive blood pressure lowering should be avoided at an early stage of ischaemic stroke.[86–88] Recent trials have investigated minimally invasive surgery and craniectomy for the treatment of patients with traumatic brain injury and spontaneous intracerebral haemorrhage (appendix pp 41–45).

Prevention and treatment of severe stroke

Approximately one fifth of patients with acute ischaemic stroke present with severe neurological deficits and 70% of patients had death or disability at either 3 months or 1 year.[89] It is important to prevent mild stroke from progressing to severe stroke, and to avoid that a severe stroke becomes a life threatening event. Recent trials have showed benefits of either dual antiplatelets or

	ChinaQUEST[62,63]	China PROGRESS[64]			CNSR I[65–67]	CNSR II[65,66]	CNSR III[66,68,69]	CSCA[70–72]	BOSC[73]
	2006	2005	2010	2015	2007–08	2012–13	2015–18	2015–22	2019–20
Participants	4783 patients; 62 hospitals[62]	7494 patients; 189 hospitals	9989 patients; 189 hospitals	10 794 patients; 189 hospitals	12 173 patients; 131 hospitals[65]	19 604 patients; 219 hospitals[65]	14 146 patients; 201 hospitals[68]	1 583 271 patients; 1930 hospitals	3 408 704 patients; 1724 hospitals
Tertiary hospitals (% of all hospitals)	48 (77·4%) of 62	..	..	..	113 (86·3%) of 131[66]	164 (74·9%) of 219[66]	163 (81·1%) of 201[68]	756 (51·2%) of 1476[71]	938 (54·4%) of 1724
Admission time (onset to door)	<3 h: 1019 (21·0%) of 4783	<24 h: 2491 (33·0%) of 7494	<24 h: 2906 (29·0%) of 9989	<24 h: 2873 (27·0%) of 10 794	<2 h: 963 (8·8%) of 10 968;[66] <3 h: 2514 (22·0%)[67] of 11 675 patients with known time	<2 h: 2406 (13·5%) of 17 823[66]	<2 h: 1715 (12·8%) of 13 397[66]	<4·5 h: 236 303 (28·0%) of 838 229;[70] <6 h: 277 653 (33·0%) of 838 229[70]	..
Access to CT	4267 (89·0%) of 4782[63]	5334 (78·7%) of 6777 for CT or MRI	7495 (81·5%) of 9192 for CT or MRI	8392 (83·7%) of 10 024 for CT or MRI	131 (99·2%) of 132 hospitals[67]	..	100% of 201 hospitals[68]	..	..
Evaluation of brain arteries	..	..	..	..	6339 (54·3%) of 11 677 eligible patients	13 239 (68·3%) of 19 382 eligible patients	195 (97·0%) of 201 hospitals[68]	1 430 639 (90·3%) of 1 583 271 eligible patients	..
Stroke unit care	1449 (30·0%) of 4782[63]	..	..	..	69 (52·4%) of 132 hospitals[67]	..	155 (77·1%) of 201 hospitals[68]	..	..
Intravenous thrombolysis	91 (2·0%) of 4783 patients	16 (0·4%) of 3579 patients admitted for <1 day	2 (0·1%) of 4374 patients admitted for <1 day	39 (0·9%) of 4579 patients admitted for <1 day	172 (1·6%) of 10 968 patients admitted for <7 days;[66] 172 (13·5%) of 1271 patients admitted for <3·5 h[66]	262 (1·5%) of 17 823 patients admitted for 7 days;[66] 262 (7·1%) of 3684 patients admitted for <3·5 h[66]	932 (7·0%) of 13 397 patients admitted for <7 days;[66] 932 (33·4%) of 2787 patients admitted for <3·5 h[66]	103 673 (6·5%) of 1 583 271 patients; 103 673 (27·2%) of 380 659 eligible patients	192 247 (5·6%) of 3 408 704 patients
Door to needle time	..	.	..	..	≤1 h: 46 (26·7%) of 172;[66] DNT: 94 (60–130) min[66]	≤1 h: 35 (13·4%) of 262;[66] DNT: 95 (72–112) min[66]	≤1 h: 498 (53·4%) of 932;[66] DNT: 60 (38–84·.5) min[66]	<1 h: 20 116 (47·5%) of 42 381[70]	≤1 h: 145 004 (75·4%) of 192 247
Thrombectomy	..	..	..	..	..	..	..	20 395 (1·3%) of 1 583 271 patients; 5197 (1·9%) of 277 653 patients admitted for <6 h[70]	49 551 (1·5%) of 3 408 704 patients
Door to puncture time	..	..	..	..	..	..	..	<1·5 h: 1484 (28·4%) of 5224[70]	≤1·5 h: 20 087 (40·5%) of 49 551
Antiplatelet therapy	In hospital: 3853 (81·0%) of 4783	In hospital: 4457 (63·6%) of 7011 eligible patients	In hospital: 6966 (73·4%) of 9490 eligible patients	In hospital: 8124 (78·5%) of 10 345 eligible patients	<48 h: 9706 (80·0%) of 12 090 eligible patients; at discharge: 8285 (71·0%) of 11 677 eligible patients	<48 h: 16 149 (85·0%) of 19 093 eligible patients; at discharge: 16 021 (90·0%) of 17 743 eligible patients	In hospital: 97·4% of eligible patients (stroke or TIA);[69] at discharge: 94·7% of eligible patients (stroke of TIA)[69]	<48 h: 1 342 678 (86·4%) of 1 554 882 eligible patients; at discharge: 1 344 784 (89·0%) of 1 511 131 eligible patients	..
Lipid-lowering drugs	In hospital: 1444 (30·0%) of 4783	In hospital: 885 (12·8%) of 6912 eligible patients	In hospital: 3599 (38·8%) of 9283 eligible patients	In hospital: 6727 (68·2%) of 9858 eligible patients	At discharge: 2402 (43·0%) of 5634 eligible patients	At discharge: 8475 (66·0%) of 12 791 eligible patients	In hospital: 96·8% of eligible patients (stroke or TIA);[69] at discharge: 93·1% of eligible patients (stroke or TIA)[69]	At discharge: 1 398 067 (91·0%) of 1 535 721 eligible patients	..

(Table 1 continues on next page)

	ChinaQUEST[62,63]	China PROGRESS[64]			CNSR I[65–67]	CNSR II[65,66]	CNSR III[66,68,69]	CSCA[70–72]	BOSC[73]
	2006	2005	2010	2015	2007–08	2012–13	2015–18	2015–22	2019–20
(Continued from previous page)									
Antihypertension therapy	In hospital: 2945 (62·0%) of 4782[63]	In hospital: 3686 (73·3%) of 5031 eligible patients	In hospital: 5150 (72·1%) of 7140 eligible patients	In hospital: 5132 (66·9%) of 7670 eligible patients	At discharge: 4620 (56·0%) of 8196 eligible patients	At discharge: 9684 (66·0%) of 14 758 eligible patients	In hospital: 60·4% of eligible patients (stroke or TIA);[69] at discharge: 64·2% of eligible patients (stroke or TIA)[69]	At discharge: 779 089 (66·3%) of 1 175 593 eligible patients	..
Anti-diabetes medication (% eligible patients)	..	In hospital: 968 (69·1%) of 1400	In hospital: 1460 (73·4%) of 1990	In hospital: 1736 (70·8%) of 2453	At discharge: 1984 (63·0%) of 3159	At discharge: 3977 (74·0%) of 5371	In hospital: 80·4% of eligible patients (stroke or TIA);[69] at discharge: 77·2% of eligible patients (stroke or TIA)[69]	At discharge: 341 304 (79·0%) of 432 145	..
Anticoagulation therapy (% eligible patients)	37 (12·0%) of 314[63]	In hospital: 21 (4·6%) of 461	In hospital: 32 (6·0%) of 537	In hospital: 62 (9·6%) of 649	At discharge: 221 (20·0%) of 1124	At discharge: 332 (21·0%) of 1578	In hospital: 28·6% of eligible patients (stroke or TIA);[69] at discharge: 34·1% of eligible patients (stroke or TIA)[69]	At discharge: 46 536 (45·7%) of 101 786	..
Neuroprotective agents	In hospital: 3632 (76·0%) of 4783	..	..	..	..	..	..	..	..
Intravenous diuretics or dehydration	In hospital: 1738 (36·0%) of 4782[63]	..	..	..	..	..	..	..	..
Traditional Chinese medicine	In hospital: 3784 (79·0%) of 4783[63]	In hospital: 5500 (78·3%) of 7028 patients	In hospital: 8297 (87·4%) of 9498 patients	In hospital: 8716 (84·0%) of 10 374 patients	..	..	..	..	..
In-hospital rehabilitation	1331 (28·0%) of 4782[63]	364 (5·5%) of 6561 eligible patients	656 (7·3%) of 9019 eligible patients	847 (8·5%) of 9915 eligible patients	5757 (49·0%) of 11 677 eligible patients	11 396 (59·0%) of 19 382 eligible patients	199 (99·9%) of 201 hospitals[68]	1 156 130 (73·0%) of 1 583 271 patients	..
Deep vein thrombosis prophylaxis	..	..	..	..	2769 (60·0%) of 4643 eligible patients	3556 (65·0%) of 5474 eligible patients	..	82 374 (17·2%) of 478 461 eligible patients	..
Dysphagia screening	..	In hospital: 180 (2·6%) of 7028 eligible patients	In hospital: 472 (5·0%) of 9497 eligible patients	In hospital: 683 (6·6%) of 10 373 eligible patients	In hospital: 4406 (36·4%) of 12 090 eligible patients	In hospital: 15 360 (83·3%) of 18 444 eligible patients	..	1 279 284 (80·8%) of 1 583 271 patients	..
Stroke education	..	..	..	..	7864 (67·3%) of 11 677 eligible patients	18 936 (97·7%) of 19 382 eligible patients	..	..	..
Smoking cessation education	..	..	..	..	2973 (63·3%) of 4694 eligible patients	7379 (85·8%) of 8600 eligible patients	..	..	..

BOSC=Bigdata Observatory platform for Stroke of China. CNSR III=the Third China National Stroke Registry. CSCA=China Stroke Center Alliance. China PROGRESS=China Patient-centered Retrospective Observation of Guideline-Recommended Performance for Stroke Sufferers. ChinaQUEST=the China Quality Evaluation of Stroke care and Treatment. DSA=digital subtraction angiography. TIA=transient ischaemic attack.

Table 1: **Nationwide registries for monitoring quality of in-hospital management of patients with ischaemic stroke in China**

Registry	Enrolment year	Setting	Follow-up method	Participants (N)	Age (years)	Admission NIHSS score	Outcomes at hospital discharge or 1 month after symptoms onset			3 months after onset of symptoms			1 year after onset of symptoms		
							Death, n (%)	Disability, n (%)	Recurrence, n (%)	Death, n (%)	Disability, n (%)	Recurrence, n (%)	Death, n (%)	Disability, n (%)	Recurrence, n (%)
Ischaemic stroke															
Chengdu Stroke Registry[94]	2002–06	Hospital-based	Medical records or telephone interview	1736	65·2±13·1	7·8±8·9	··	··	··	154 of 1612 (9·6%)	537 of 1458 (36·8%)	··	228 of 1438 (15·9%)	281 of 1210 (23·2%)	··
Chengdu Stroke Registry[94]	2007–11	Hospital-based	Medical records or telephone interview	2335	63·9±13·6	6·8±6·7	··	··	··	175 of 1918 (9·1%)	465 of 1743 (26·7%)	··	226 of 1883 (12%)	332 of 1657 (20%)	··
Chengdu Stroke Registry[94]	2012–16	Hospital-based	Medical records or telephone interview	2391	63·8±14·1	6·7±6·4	··	··	··	133 of 2078 (6·4%)	558 of 1945 (28·7%)	··	197 of 1842 (10·7%)	289 of 1645 (17·6%)	··
ChinaQUEST[63,95]	2006	Hospital-based	Medical records or telephone interview	4782	65±12	··	151 of 4782 (3·1%)[63]	1606 of 4782 (33·6%)[63]	··	··	··	··	491 of 4782 (10·3%)[95]	942 of 4782 (19·7%)[95]	··
CNSR I[65,96,97]	2007–08	Hospital-based	Telephone interview or local citizen registry	12 173	67 (57–75)	··	496 of 12 173 (4·1%)[65]	3487 of 12 173 (28·6%)[65]	242 of 9314 (2·6%)[96]	1037 of 11 550 (9·0%)[97]	2952 of 10 523 (28·1%)[97]	1494 of 11 560 (12·9%)[97]	1664 of 11 560 (14·4%)[97]	2241 of 9896 (22·6%)[97]	2050 of 11 560 (17·7%)[97]
CNSR II[65, 98]	2012–13	Hospital-based	Telephone interview	19 604	65 (57–74)	··	222 of 19 604 (1·1%)[65]	4670 of 19 604 (23·9%)[65]	··	··	··	··	1645 of 19 341 (8·5%)[98]	5046 of 17 340 (29·1%)[98]	1289 of 19 341 (6·7%)[98]
CNSR III[99]	2015–18	Hospital-based	Face-to-face or telephone interview	9038	62·4±11·4	4·6±4·3	··	··	··	136 of 9038 (1·5%)	1306 of 8930 (14·6%)	591 of 9038 (6·5%)	307 of 9038 (3·4%)	1225 of 8818 (13·9%)	930 of 9038 (10·3%)
China PROGRESS[64]	2005	Hospital-based	Medical records	7494	67 (57–74)	··	133 of 7324 (1·8%)	··	··	··	··	··	··	··	··
China PROGRESS[64]	2010	Hospital-based	Medical records	9989	68 (59–75)	··	74 of 9796 (0·8%)	··	··	··	··	··	··	··	··
China PROGRESS[64]	2015	Hospital-based	Medical records	10794	67 (59–76)	··	67 of 10 538 (0·6%)	··	··	··	··	··	··	··	··
CSCA[72]	2015–22	Hospital-based	Medical records	1 583 271	66·3±12	··	39 028 of 1 367 585 (2·9%)	321 141 of 981 050 (32·7%)	··	··	··	··	··	··	··
BOSC	2018[14]	Hospital-based	Telephone interview	74 904	68 (58–74)	3 (1–5)	924 of 74 904 (1·2%)	··	··	3015 of 68 742 (4·4%)	··	2912 of 68 742 (4·2%)	4680 of 68 742 (6·8%)	9879 of 64 062 (15·4%)	5322 of 68 742 (7·7%)
BOSC	2019[100]	Hospital-based	Telephone interview	27 316	66 (57–74)	3 (1–5)	256 of 27 316 (0·9%)	··	··	··	··	··	1514 of 25 130 (6·0%)	3363 of 23 616 (14·2%)	1500 of 23 616 (6·4%)
CKB[101]*	2004–08	Community-based	Death registries or medical records	36 588	59·3±9·6	··	1098 of 36 588 (3·0%)	··	··	··	··	··	1169 of 30 206 (3·9%)	··	4708 of 27 126 (17·4%)

(Table 2 continues on next page)

Registry	Enrolment year	Setting	Follow-up method	Participants (N)	Age (years)	Admission NIHSS score	Outcomes at hospital discharge or 1 month after symptoms onset			3 months after onset of symptoms			1 year after onset of symptoms		
							Death, n (%)	Disability, n (%)	Recurrence, n (%)	Death, n (%)	Disability, n (%)	Recurrence, n (%)	Death, n (%)	Disability, n (%)	Recurrence, n (%)
(Continued from previous page)															
Intracerebral haemorrhage															
ChinaQUEST[95,102]	2006	Hospital-based	Medical records or telephone interview	1572	61±13	..	201 of 1572 (12·8%)[102]	802 of 1572 (51·0%)[102]	..	..	..	..	415 of 1572 (26·4%)[95]	369 of 1572 (23·5%)[95]	..
CNSR I[96,103]	2007–08	Hospital-based	Medical records or telephone interview	3828	60±14	9 (3–16)	325 of 3497 (9·3%)[96]	..	66 of 3497 (1·9%)[96]	651 of 3255 (20·0%)[103]	947 of 3255 (29·1%)[103]	..	850 of 3255 (26·1%)[103]	648 of 3255 (19·9%)[103]	..
CNSR II[98]	2012–13	Hospital-based	Telephone interview	3042	..	..	..	..	..	..	..	..	507 of 3042 (16·7%)	1259 of 2668 (47·2%)	217 of 3042 (7·1%)
CSCA[71]	2015–19	Hospital-based	Medical records	85 705	62·9±12·9	..	2017 of 85 705 (2·4%)	..	..	..	..	..	..	..	..
BOSC[100]	2019	Hospital-based	Telephone interview	7174	62 (53–71)	..	364 of 7174 (5·1%)	..	..	..	..	..	1088 of 6130 (17·7%)	1471 of 5042 (29·2%)	188 of 5042 (3·7%)
CKB[101]*	2004–08	Community-based	Death registries or medical records	7440	59·6±10·2	..	3497 of 7440 (47·0%)	..	..	..	..	..	402 of 3575 (11·2%)	..	741 of 3010 (24·6%)
Subarachnoid haemorrhage															
CNSR I[96]	2007–08	Hospital-based	Medical records or telephone interview	629	55·9±12·8	0 (0–2)	56 of 557 (10·1%)	..	40 of 557 (7·2%)	..	..	..	..	..	..
CSCA[71]	2015–19	Hospital-based	Medical records	11 241	60·0±12·9	..	341 of 11 241 (3·0%)	..	..	..	..	..	..	..	..
BOSC[100]	2019	Hospital-based	Telephone interview	1760	59 (51–68)	..	65 of 1760 (3·7%)	..	..	..	..	..	214 of 1531 (14·0%)	146 of 1317 (11·1%)	33 of 1317 (2·5%)
CKB[101]*	2004–08	Community-based	Death registries or medical records	702	56±10	..	133 of 702 (19·0%)	..	..	..	..	..	..	..	..

The table includes both hospital-based and community-based studies in China, where some patients enrolled in community-based studies might not have been treated for stroke. The differences in recurrent stroke rates between community-based and hospital-based studies might reflect differences in study population, sampling methods and time periods, and approaches and completeness of ascertaining recurrent events. In hospital-based studies, recurrent events were collected primarily via telephone interviews as self-reports by patients or families. In community-based studies, recurrence events were collected from electronic record linkage with health insurance systems and mortality registrars. BOSC=Bigdata Observatory platform for Stroke of China. CKB=China Kadoorie Biobank. CNSR=China National Stroke Registry. CSCA=Chinese Stroke Centre Alliance. NIHSS=National Institutes of Health Stroke Scale. *The CKB study reported that, in patients with stroke who survived for 28 days after onset of symptoms, the cumulative recurrence rate for ischaemic stroke was 41% at 5 years and 54% at 9 years, and the cumulative mortality rate was 16% at 5 years and 29% at 9 years; the cumulative recurrence rates after intracerebral haemorrhage were 44% and 53%, respectively, and the cumulative mortality rates were 28% and 41% at 5 years and 9 years, respectively; the cumulative recurrence rate was 22% and the cumulative mortality rate was 16% at 5 years after subarachnoid haemorrhage.

Table 2: **Outcomes of patients with stroke in China**

intravenous tirofiban, versus aspirin alone, in preventing early neurological deterioration. In patients with stroke and early neurological deterioration, either intravenous tirofiban or argatroban improved functional outcomes compared with conventional antiplatelet therapy (appendix pp 24–40). However, there is scarce evidence to inform optimal management for patients with severe stroke, as they are often excluded from clinical trials. The Chinese Stroke Society issued a clinical guideline in 2024 to outline the neurocritical care and specialised neurological interventions for patients with severe stroke, based on evidence that was mostly derived from studies of general stroke cohorts.[76] Chinese researchers have started to focus their attention on severe stroke, with studies undertaken to clarify its definition,[90] predict neurological complications,[91] and optimise nutritional regimens.[92]

Complementary, neuroprotective and other therapies

Traditional Chinese medicine is widely used for the treatment of patients with stroke in China (table 1). The number of well-designed trials to evaluate the role of specific herbal compounds and acupuncture in stroke patients is increasing. There is scope for further research on how traditional Chinese medicine and Western medicine might interact, and to explore their potential synergistic or antagonistic effects.[93] Some neuroprotective agents with anti-inflammatory and microcirculation protective effects have been shown to improve functional outcomes in some patients with stroke; furthermore, non-pharmacological interventions, such as normobaric hyperoxia and remote ischaemic conditioning have promising clinical implications (appendix pp 46–54).

Outcome and recovery after stroke

In the past decade, the considerable progress in stroke management has improved patients' outcomes, as indicated by the decrease in hospital deaths, recurrence rate, and death or disability at 3 months and 1 year, for both ischaemic and haemorrhagic strokes (table 2).[14,63–65,72,94–103]

Secondary prevention

The improvement in secondary prevention might be attributed to the improved control of vascular risk factors. On the one hand, the use of secondary prevention medications has increased in hospital (table 1). On the other hand, the national centralised drug procurement scheme includes all essential medications for secondary stroke prevention, such as aspirin and statins, and medications for hypertension and diabetes. The scheme ensures equitable access to medications with affordable prices.[104] However, adherence to evidence-based medications for secondary prevention decreases after hospital discharge, particularly for antithrombotic agents, statins, and antidiabetics.[69]

Given the diversity in pathology and genetic background of patients, researchers in China have been working on individualised strategies for secondary prevention. Aspirin is the established treatment for secondary prevention of atherosclerotic stroke.[105] For patients with minor stroke or high-risk transient ischaemic attack, it is an established strategy to commence with a combination of aspirin and clopidogrel within 24 h after stroke onset and then switch to a single antiplatelet agent after 21 days.[75] A recent trial has broadened the definition of minor stroke, with an upper limit of scoring from 3–5 on the National Institutes of Health Stroke Scale, and extended the initiation time of dual antiplatelet therapy from within 24 h to 72 h after symptoms onset.[106] Clopidogrel resistance is associated with *CYP2C19* loss-of-function alleles. About 60% of Asian people, compared with 25% of Western people, are carriers of these alleles.[107] The point-of-care testing offers a genotype-guided antiplatelet approach at the hospital bedside, by which ticagrelor has been shown more effective than clopidogrel combined with aspirin in reducing stroke recurrence for carriers of *CYP2C19* loss-of-function alleles.[108] Cilostazol is another antithrombotic agent shown to be safe and effective in secondary stroke prevention when used either as monotherapy or combination therapy with aspirin or clopidogrel, compared with placebo or best medical therapy.[109] A recent trial implied that, for patients at high risk of postoperative ischaemic events, early initiation of aspirin therapy after surgery for intracerebral haemorrhage reduced the risk of major ischaemic events without any increase in the risk of intracranial haemorrhage.[110]

Rehabilitation

Stroke rehabilitation in China integrates a range of interventions involving physical therapy, occupational therapy, psychotherapy, and traditional Chinese therapies.[111] In-hospital rehabilitation is now available for most patients, and more than 70% of patients with stroke receive a rehabilitation assessment while hospitalised (table 1). In 2021, the Chinese government issued policies to improve the capacity of medical rehabilitation services by the creation of rehabilitation hospitals, along with rehabilitation departments in general hospitals.[112] Novel technologies that involve the use of telemedicine appear to be cost-effective for self-directed or remote-supervised rehabilitation in the community.[113]

Emerging research frontiers for stroke in China

In the past decade, China has made great achievements regarding evidence-based clinical practice for the management of stroke. There has been a surge of rigorous randomised trials (appendix pp 16–17), particularly for reperfusion therapies and secondary prevention after ischaemic stroke, which provide robust

Panel 1: Summary findings of genetic studies of stroke in China

Mendelian randomisation studies for primary prevention
- Moderate intake of alcohol consumption significantly increased risk of both ischaemic stroke and intracerebral haemorrhage[37]
- Elevation in systolic blood pressure increased risk of both ischaemic stroke and intracerebral haemorrhage[114]
- High levels of homocysteine increased risks of both ischaemic stroke and intracerebral haemorrhage[115]
- High levels of LDL cholesterol increased risk of ischaemic stroke and decreased risk of intracerebral haemorrhage[116]
- Plasma lipoprotein(a) levels increased risk of large artery ischaemic stroke[117]
- Snoring causally increased risk of both ischaemic and haemorrhagic stroke[118]

Secondary prevention
- Ticagrelor plus aspirin reduced stroke recurrence compared with clopidogrel plus aspirin in carriers of *CYP2C19* loss-of-function alleles[108]
- Genotype-guided warfarin dosing was superior to standard clinical dosing in maintaining the international normalised ratio[119]

Stroke-omics atlas
- Genomic data revealed genetic susceptibility signals and potential novel therapeutic targets across stroke subtypes
- Loss-of-function variants in *NOTCH3* displayed a broad neuroimaging spectrum, such as imaging markers of cerebral small vessel diseases in patients with CADASIL[120]
- Integrative genomic and proteomic analyses identified novel drug targets for stroke, such as F11, ALDH2, and MMP12[121]
- An integrated polygenic risk score can predict the risk of first-ever stroke and recurrent stroke[122]

Panel 2: Priorities for stroke research and improvement of clinical management of stroke patients in China

Control major risk factors
Collaboration from the government, industries, health professionals, and individuals to:
- Provide more intensive monitoring of blood pressure and blood concentrations of lipid and glucose in people at high risk of stroke
- Encourage a healthy diet with low sodium intake and high fruit and vegetable consumption
- Further control of tobacco and alcohol consumption and air pollution, even at the cost of short-term commercial profit

Improve the capacity and quality of health services
Governmental policies and investment to:
- Refine pre-hospital emergency medical systems for stroke
- Provide public education and doctor training for better adherence to evidence-based stroke care
- Strengthen long-term rehabilitation and secondary prevention

Enlarge the scale of health industries
Implement well-designed trials to:
- Examine and refine individualised treatment based on stroke aetiologies and severity
- Further assess the safety and effectiveness of widely used traditional Chinese medicine and neuroprotective agents
- Investigate cost-effective strategies for stroke prevention and treatment in low-resource setting, such as rural areas
- Explore novel approaches, such as genetics and artificial intelligence to inform personalised medicine for stroke

Upgrade health systems and policies
- Nationwide survey for up-to-date stroke epidemiology to optimise allocation of health-care resources
- Motivation and training of primary care physicians and village doctors to act as sentinel points for stroke care
- Governmental action on improved infrastructure and public education for stroke prevention, early recognition of stroke, and timely access to stroke services

Enhance overall health promotion
Public-health awareness and education throughout the whole lifecycle:
- Start with schoolchildren for a healthy lifestyle
- Target adult men to stop smoking and cease alcohol consumption
- Motivate older people to get screened for and control vascular risk factors
- Encourage suspected patients with stroke to use emergency medical services to ensure timely access to stroke care

evidence to refine Chinese and international stroke guidelines. For intracerebral haemorrhage, there is potential for wider benefit by implementation of care bundle management, in coordination with surgery and other medical treatment. Novel technologies have advanced personalised medicine approaches for stroke care. For example, stroke genetics have advanced our understanding of stroke aetiology, enhanced primary prevention, and informed individualised secondary prevention (panel 1).[37,108,114–122] Moreover, artificial intelligence would facilitate more efficient and accurate prediction and diagnosis of stroke.[123] Several studies are ongoing that will provide high-quality evidence to guide stroke management (appendix pp 16–17).

Search strategy and selection criteria

We searched MEDLINE and Chinese National Knowledge Infrastructure for articles published in English or Chinese between November, 2018 and August, 2025 for the MeSH and free text search terms: ("stroke" OR "ischaemic stroke" OR "cerebral infarction" OR "intracerebral haemorrhage" OR "intracranial haemorrhage" OR "cerebral haemorrhage" OR "subarachnoid haemorrhage") AND ("epidemiology" OR "epidemiological" OR "epidemiologic" OR "epidemic" OR "burden" OR "incidence" OR "prevalence" OR "mortality" OR "death" OR "fatality" OR "disability adjusted life year" OR "risk factors" OR "primary prevention" OR "secondary prevention" OR "diagnosis" OR "treatment" OR "intervention" OR "therapy" OR "management" OR "stroke care" OR "rehabilitation" OR "recovery" OR "outcome" OR "prognosis") AND ("China"). The search was restricted to human studies. We included studies conducted or initiated in China that reported on the epidemiology, clinical characteristics and outcomes, management and patterns of clinical practice, primary and secondary prevention, therapeutic trials, national strategies for care, cultural issues affecting clinical practice, and guidelines for prevention and management of stroke. We also inspected the reference lists of the retrieved articles and reviews, consulted experts in the field, and searched ClinicalTrials.gov and the Chinese Clinical Trial Registry. Additionally, we reviewed websites of the Global Burden of Disease Study, the State Council, the National Health Commission, and the National Bureau of Statistics of the People's Republic of China for relevant national policies and government reports in December, 2025. The final reference list was generated according to relevance and quality. Search strategies and selection criteria are detailed in the appendix (pp 8–11).

Conclusion

China has made great advances in stroke care and research in the past decade, particularly in acute treatment and secondary prevention. Progress continues in prevention, diagnosis, and management of stroke, although gaps remain in clinical practice and research (panel 2). With a national vision to construct a healthy China (figure 3), multidisciplinary collaborations at both national and individual levels (ie, by individual resident or patient) are needed to further reduce the burden of stroke. Future efforts should focus on promoting healthy lifestyle, public education on awareness of stroke prevention and timely access to stroke services, training of stroke clinicians for better adherence to evidence-based stroke care, and improvements in both pre-hospital and post-hospital services. By providing wider access to full lifecycle health-care services for stroke prevention and care, China can achieve the ultimate goal of improving the overall quality of life for Chinese people.

Contributors

SW, BW, and ML conceived the work. SW, YaW, and JL drafted the outline of the work and CSA, ZC, PAGS, TNN, BW, and ML advised on the outline. SW, YaW, and JL conducted the literature search, evaluated the quality of the included studies, and performed data collection and analysis. SW, YaW, and JL wrote the first draft. SW, BW, and ML revised each version of the manuscript. All authors sourced data from key publications in their areas of expertise, interpreted data, and revised the manuscript following major changes. All authors added extra information and thoroughly edited the manuscript. All authors read and approved the final version of the manuscript.

Declaration of interests

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